

Data Warehousing MCQs – Correct Answers and Evaluation

1. Which of the following is NOT a characteristic of a Data Warehouse?

- a) Integrated
- b) Time-variant
- c) **Normalized** ☒ (Correct Answer: c)
- d) Subject-oriented

✗ Your Answer: b

A data warehouse is characterized by being subject-oriented, integrated, non-volatile, and time-variant.

2. The process of transforming raw data into a format that can be analyzed is called:

- a) ETL
- b) **OLAP** ☒ (Correct Answer: b)
- c) Data Mining
- d) Metadata Management

☒ Your Answer: b

3. What is the main purpose of OLAP in Data Warehousing?

- a) Execute transactional queries
- b) **Support real-time analytics** ☒ (Correct Answer: b)
- c) Optimize normalized tables
- d) Store structured and unstructured data

☒ Your Answer: b

4. Which type of schema is most normalized?

- a) Star Schema
- b) **Snowflake Schema** ☒ (Correct Answer: b)
- c) Fact Constellation Schema
- d) Hybrid Schema

✗ Your Answer: c

5. In a Star Schema, the fact table contains:

- a) Only primary keys
- b) Only numerical values
- c) **Both foreign keys and measures** ☒ (Correct Answer: c)
- d) Only dimension attributes

✗ Your Answer: b

6. Which of the following best defines the concept of a slowly changing dimension (SCD)?

- a) A dimension table that never changes
- b) A dimension that changes frequently
- c) **A dimension whose attributes change over time** ☒ (Correct Answer: c)
- d) A fact table that stores historical records

☒ Your Answer: d

7. In ETL, the process of cleansing data to remove inconsistencies is known as:

- a) **Data Scrubbing** ☒ (Correct Answer: d)
- b) Data Profiling
- c) Data Purging
- d) Transformation

☒ Your Answer: a

8. What is the purpose of a surrogate key in a Data Warehouse?

- a) To maintain referential integrity
- b) **To uniquely identify a record irrespective of source system changes** ☒ (Correct Answer: b)
- c) To normalize data
- d) To replace foreign keys in a fact table

☒ Your Answer: b

9. In a Data Warehouse, what does the "drill-through" operation allow?

- a) Viewing summarized data
- b) **Moving from a summary to detailed data** ☒ (Correct Answer: b)
- c) Aggregating multiple reports
- d) Creating new data marts

☒ Your Answer: b

10. A data mart is:

- a) **A subset of a data warehouse focused on a specific business area** ☒ (Correct Answer: a)
- b) A temporary storage area for ETL processing
- c) The same as an OLTP database
- d) A method of data cleansing

☒ Your Answer: a

11. Which of the following is NOT an advantage of Data Warehousing?

- a) Improved query performance
- b) **Real-time transaction processing** ☒ (Correct Answer: b)
- c) Historical data analysis
- d) Data integration

☒ Your Answer: b

12. What is the primary purpose of metadata in a Data Warehouse?

- a) Storing transactional data
- b) **Defining the structure and meaning of data** ☒ (Correct Answer: b)
- c) Optimizing SQL queries
- d) Storing historical logs

☒ Your Answer: b

13. Which of the following is a key difference between OLTP and OLAP?

- a) OLTP focuses on long-term data analysis, OLAP handles daily transactions
- b) OLTP has complex queries, OLAP only has simple queries
- c) OLAP is designed for real-time transactional operations, OLTP for analytics
- d) **OLAP focuses on data analysis, OLTP handles transaction processing** ☒ (Correct Answer: d)

☒ Your Answer: d

14. Which of these technologies is commonly used for implementing a Data Warehouse?

- a) PostgreSQL
- b) Hadoop
- c) MySQL
- d) **Amazon Redshift** ☒ (Correct Answer: d)

☒ Your Answer: d

15. What is the main benefit of a Fact Table in a Data Warehouse?

- a) It contains descriptive attributes of dimensions
- b) It stores aggregated data for analysis
- c) **It contains historical transactional data** ☒ (Correct Answer: c)
- d) It normalizes data to reduce redundancy

☒ Your Answer: a

16. Which of the following is NOT a component of an ETL process?

- a) Extraction
- b) **Transaction processing** ☒ (Correct Answer: b)

- c) Transformation
- d) Loading

☒ Your Answer: b

17. What does MOLAP stand for?

- a) Modern Online Analytical Processing
- b) **Multi-Dimensional Online Analytical Processing** ☒ (Correct Answer: b)
- c) Metadata Online Analytical Processing
- d) Managed OLTP and Analytical Processing

☒ Your Answer: b

18. Which of these is an example of a Slowly Changing Dimension (SCD) Type 2 implementation?

- a) Overwriting old records
- b) **Creating a new row for each change** ☒ (Correct Answer: b)
- c) Ignoring changes to dimension attributes
- d) Maintaining a backup copy of the fact table

☒ Your Answer: b

19. What is a conformed dimension in Data Warehousing?

- a) **A dimension that is consistent across multiple fact tables** ☒ (Correct Answer: a)
- b) A dimension that changes frequently
- c) A temporary lookup table for queries
- d) A separate schema for storing historical data

☒ Your Answer: a

20. What is the role of a staging area in Data Warehousing?

- a) Storing data permanently
- b) Performing OLAP queries
- c) **Temporarily storing data before ETL processing** ☒ (Correct Answer: c)
- d) Managing real-time analytics

☒ Your Answer: c

1. Which of the following is NOT a valid transformation in Informatica?

- a) Router
- b) Aggregator
- c) **Fact Lookup** ☒ (Correct Answer: c)
- d) Filter

☒ Your Answer: c

2. What is the primary purpose of the Lookup transformation in Informatica?

- a) Performing arithmetic operations
- b) Filtering rows from a dataset
- c) Joining data from multiple sources
- d) **Retrieving related data from another table** ☒ (Correct Answer: d)

☒ Your Answer: d

3. In Informatica, which transformation is used to split data into multiple pipelines based on conditions?

- a) Filter
- b) Joiner
- c) **Router** ☒ (Correct Answer: c)
- d) Normalizer

☒ Your Answer: c

4. What is the function of the 'Aggregator' transformation?

- a) Performing lookup operations
- b) Applying filters
- c) **Performing calculations such as SUM, AVG, COUNT** ☒ (Correct Answer: c)
- d) Sorting data

☒ Your Answer: c

5. Which of the following is NOT a valid mode in Lookup transformation?

- a) Connected
- b) Unconnected
- c) **Static** ☒ (Correct Answer: c)
- d) Hybrid

☒ Your Answer: b

6. Which transformation is best suited for handling hierarchical data like XML?

- a) Normalizer
- b) **XML Parser** ☒ (Correct Answer: b)
- c) Joiner
- d) Rank

☒ Your Answer: d

7. The primary function of the 'Rank' transformation is to:

- a) Filter data based on conditions
- b) **Sort and rank data based on a given column** ☒ (Correct Answer: b)
- c) Merge multiple datasets
- d) Perform complex mathematical calculations

☒ Your Answer: b

8. What does the Normalizer transformation do in Informatica?

- a) Converts multiple rows into a single row
- b) **Converts a single row into multiple rows** ☒ (Correct Answer: b)
- c) Removes duplicate records
- d) Sorts data in ascending order

☒ Your Answer: b

9. What is the purpose of the 'Update Strategy' transformation?

- a) **Handling data updates, inserts, and deletes** ☒ (Correct Answer: a)
- b) Sorting records in a dataset
- c) Filtering rows based on conditions
- d) Joining multiple datasets

☒ Your Answer: a

10. Which type of Join is NOT supported by Informatica's Joiner transformation?

- a) Normal Join
- b) Left Outer Join
- c) Right Outer Join
- d) **Full Outer Join** ☒ (Correct Answer: d)

☒ Your Answer: a

11. In Informatica, which caching mechanism improves Lookup performance?

- a) Static Cache
- b) Dynamic Cache
- c) Persistent Cache
- d) **All of the above** ☒ (Correct Answer: d)

☒ Your Answer: d

12. The 'Transaction Control' transformation allows:

- a) Error handling
- b) Conditional row processing
- c) **Performing commit and rollback operations** ☒ (Correct Answer: c)
- d) Sorting data

☒ Your Answer: c

13. What happens if a mapping contains an unconnected Lookup transformation?

- a) The transformation is ignored
- b) **It must be manually invoked using an expression** ☒ (Correct Answer: b)
- c) It processes all input records
- d) The workflow fails

☐ Your Answer: a

14. What is the function of the 'Sorter' transformation?

- a) Removing duplicates
- b) **Sorting records in ascending or descending order** ☒ (Correct Answer: b)
- c) Aggregating data
- d) Filtering rows

☒ Your Answer: b

15. Informatica PowerCenter's Session Task is used to:

- a) Define the execution logic of a workflow
- b) **Move data from source to target** ☒ (Correct Answer: b)
- c) Validate the mappings
- d) Manage repository objects

☐ Your Answer: c

16. What is the primary difference between Connected and Unconnected Lookup?

- a) **Connected Lookup can return multiple columns, while Unconnected Lookup returns a single value** ☒ (Correct Answer: a)
- b) Connected Lookup cannot be used in mappings
- c) Unconnected Lookup always executes faster
- d) Unconnected Lookup supports complex joins

☒ Your Answer: a

17. Which of these transformations is an Active transformation?

- a) **Filter** ☒ (Correct Answer: a)
- b) Expression
- c) Sequence Generator
- d) Lookup

☒ Your Answer: a

18. In Informatica, which partitioning type ensures even data distribution?

- a) **Round Robin** ☒ (Correct Answer: a)
- b) Hash Auto Key
- c) Key Range
- d) Database Partitioning

☒ Your Answer: d

19. What is the default behavior of the Joiner transformation when joining two datasets?

- a) **Normal Join** ☒ (Correct Answer: b)
- b) Left Outer Join
- c) Full Outer Join
- d) Right Outer Join

☒ Your Answer: c

20. Informatica PowerCenter's Repository contains:

- a) Only workflows
- b) **Metadata, mappings, sessions, and workflows** ☒ (Correct Answer: b)
- c) Only source and target definitions
- d) Only user-defined functions

☒ Your Answer: b

1. What is the primary purpose of a 'Data Extract' in Tableau?

- a) **To improve performance by storing a subset of data** ☒ (Correct Answer: a)
- b) To connect Tableau with live data sources
- c) To create calculated fields
- d) To refresh dashboards automatically

☒ Your Answer: a

2. What type of join is NOT available in Tableau?

- a) Inner Join
- b) Right Join

- c) **Cross Join** ☒ (Correct Answer: c)
d) Full Outer Join

☒ Your Answer: c

3. What is the function of a 'LOD (Level of Detail) Expression' in Tableau?

- a) To limit the number of rows in a dataset
b) **To aggregate data at different granularities** ☒ (Correct Answer: b)
c) To apply filters dynamically
d) To join datasets efficiently

☒ Your Answer: b

4. In Tableau, which chart is best suited to show trends over time?

- a) Bar Chart
b) Scatter Plot
c) **Line Chart** ☒ (Correct Answer: c)
d) Heatmap

☒ Your Answer: c

5. What is the purpose of a 'Dual-axis' chart?

- a) To combine two different data sources
b) **To visualize two different measures on the same chart** ☒ (Correct Answer: b)
c) To compare data from multiple dashboards
d) To filter multiple sheets simultaneously

☒ Your Answer: b

6. Which of the following is NOT a valid type of filter in Tableau?

- a) Dimension Filter
b) Measure Filter
c) **Table Filter** ☒ (Correct Answer: c)
d) Context Filter

☒ Your Answer: c

7. What does a 'Context Filter' do in Tableau?

- a) **It improves performance by pre-filtering data** ☒ (Correct Answer: a)
b) It creates a copy of the dataset
c) It allows only one measure per visualization
d) It groups multiple filters into one

☒ Your Answer: a

8. What is a 'Calculated Field' used for in Tableau?

- a) To store a snapshot of a dataset
- b) **To create new values based on existing data** ☒ (Correct Answer: b)
- c) To join two datasets
- d) To remove duplicate values

☒ Your Answer: b

9. Which file format is used to store Tableau workbooks with extracted data?

- a) .twb
- b) .tds
- c) .twbx
- d) **.tde** ☒ (Correct Answer: d)

☒ Your Answer: d

10. What is the use of a 'Set' in Tableau?

- a) **To filter data dynamically based on conditions** ☒ (Correct Answer: a)
- b) To replace calculated fields
- c) To improve performance
- d) To create new measures

☒ Your Answer: d

11. Which of the following is a difference between 'Extract' and 'Live' connection in Tableau?

- a) Extracts require an internet connection, but Live does not
- b) **Extracts allow faster performance, while Live connects to real-time data** ☒ (Correct Answer: b)
- c) Live connections store data locally
- d) Extracts are always slower than Live connections

☒ Your Answer: b

12. What is the main use of 'Bins' in Tableau?

- a) To create custom date ranges
- b) **To create equal-sized groups from a continuous field** ☒ (Correct Answer: b)
- c) To filter dimensions
- d) To aggregate data

☒ Your Answer: b

13. Which calculation type is NOT available in Tableau?

- a) Table Calculation
- b) Row-Level Calculation
- c) Aggregate Calculation
- d) **Index Calculation** ☒ (Correct Answer: d)

✗ Your Answer: b

14. What does a 'Hierarchy' allow users to do in Tableau?

- a) Automatically sort data
- b) **Drill down from a higher-level category to a lower-level category** ☒ (Correct Answer: b)
- c) Remove duplicate values
- d) Create advanced calculations

☒ Your Answer: b

15. What is 'Blending' in Tableau?

- a) Joining two datasets within the same data source
- b) **Combining data from multiple sources without merging** ☒ (Correct Answer: b)
- c) Creating duplicate data fields
- d) Using multiple dashboards in one workbook

☒ Your Answer: b

16. What type of chart is best for showing proportions?

- a) Line Chart
- b) **Pie Chart** ☒ (Correct Answer: b)
- c) Scatter Plot
- d) Histogram

✗ Your Answer: c

17. In Tableau, what does the 'Pages' shelf do?

- a) **Divides the visualization into multiple pages based on a field** ☒ (Correct Answer: a)
- b) Creates new worksheets
- c) Adds filters to dashboards
- d) Duplicates sheets for different data sources

☒ Your Answer: a

18. What is 'Story' used for in Tableau?

- a) **To create a sequence of visualizations that convey a message** ☒ (Correct Answer: a)
- b) To blend data sources
- c) To join multiple dashboards
- d) To apply global filters

☒ Your Answer: a

19. What is the use of the 'Forecast' feature in Tableau?

- a) **To predict future trends based on historical data** ☒ (Correct Answer: a)
- b) To filter data dynamically
- c) To compare two different datasets
- d) To add annotations

☒ Your Answer: a

20. Which option is NOT available in Tableau's 'Dashboard Actions'?

- a) Filter Action
- b) Highlight Action
- c) Parameter Action
- d) **Row-Level Action** ☒ (Correct Answer: d)

☒ Your Answer: d